Air Pollution



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• *The first primitive living cells could not have occurred or continued to exist in an atmosphere composed of the chemical constituents we now regard as normal and requisite to life.*

Introduction

- Most eminent problems we face – Environmental pollution
  - Although environmental pollution can be caused by natural events such as forest fires and active volcanoes, use of the word pollution generally implies that the contaminants have an anthropogenic source—that is, a source created by human activities.
  - From the beginning of the fourteenth century to the early part of the twentieth, air pollution by coal smoke and gases occupied the center of the stage almost exclusively, and in many industrialized areas of the world it is still the dominant concern.
  - By the middle of the 20th century, an awareness of the need to protect air, water, and land environments from pollution had developed among the general public.
  - Major pieces of environmental legislation, such as the Clean Air Act (1970) and the Clean Water Act (1972; United States), were passed in many countries to control and mitigate environmental pollution.







Introduction

- Air Pollution definition:

"Air pollution means the presence in the outdoor atmosphere of one or more contaminants, such as dust, fumes, gas, mist, odour, smoke, or vapour, in quantities, with characteristics, and of durations such as to be injurious to human, plant, or animal life or to property, or which unreasonably interfere with the comfortable enjoyment of life and property."

—Engineers Joint Council (U.S.A.)

"Air pollution is the presence in ambient atmosphere of substances, generally resulting from the activity of man, in sufficient concentration, present for a sufficient time and under circumstances which interfere significantly with the comfort, health or welfare of persons or with the full use or enjoyment of property."

—Indian Standards Institution IS-4167 (1966).

"Air pollution is the excessive concentration of foreign matter in the air which adversely affects the well being of the individual or causes damage to property."

—American Medical Association

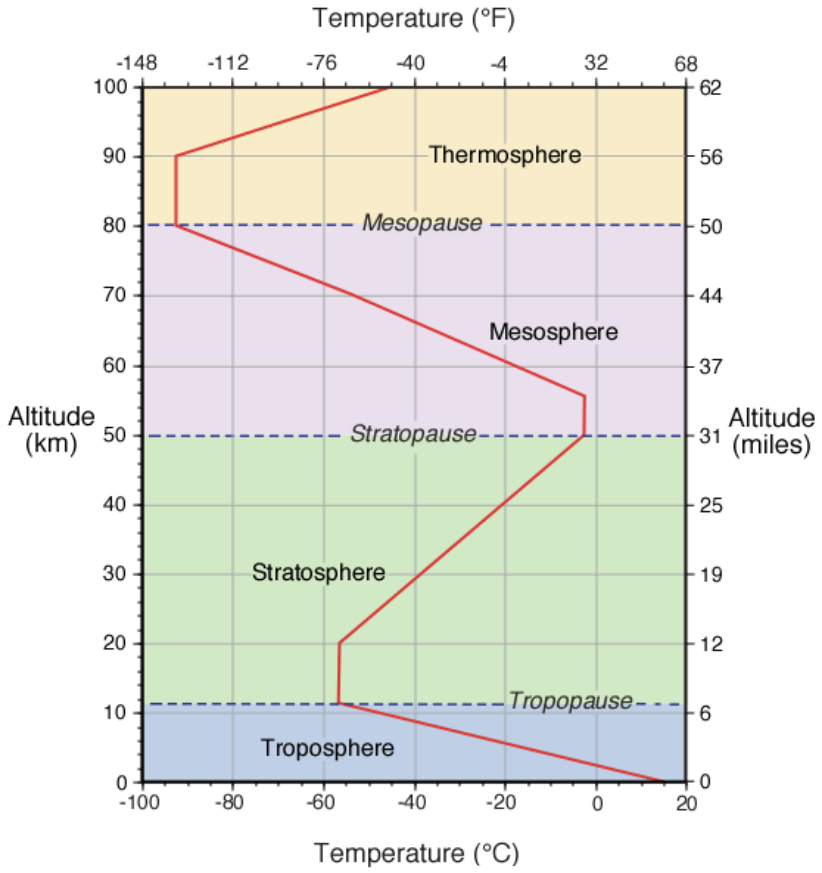
Air pollution is contamination of the indoor or outdoor environment by any chemical, physical or biological agent that modifies the natural characteristics of the atmosphere.

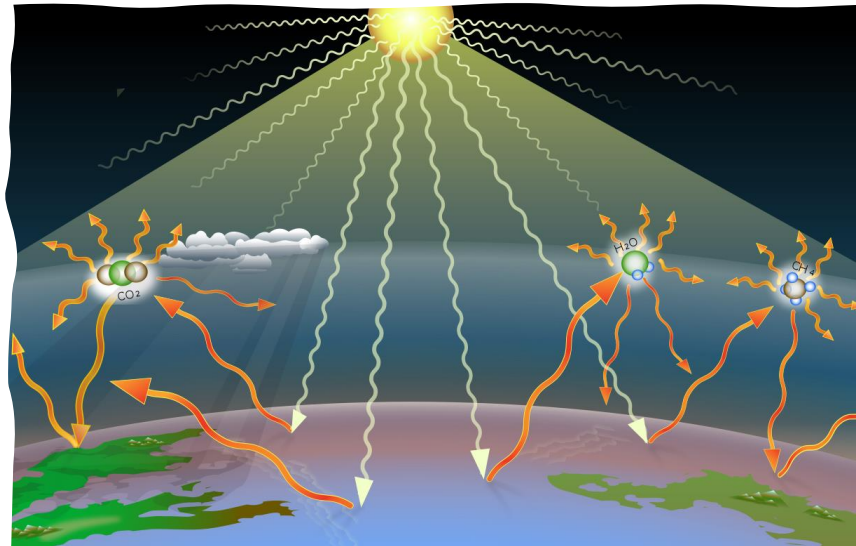
-WHO

WHO data show that almost all of the global population (99%) breathe air that exceeds WHO guideline limits and contains high levels of pollutants, with low- and middle-income countries suffering from the highest exposures.

Introduction

- Human and Atmosphere
  - Greenhouse gas-induced warming in the troposphere is causing it to expand and squash the stratosphere. (CO_2, Water Vapor, Methane, Nitrous Oxide)
  - Carbon dioxide in the atmosphere have increased by 38 percent since 1750, while methane concentrations have gone up 148 percent.
  - Acid Rain: Acidification of soil and lakes
  - Depleted Ozone Layer: CFC, Aerosol





The 28th Conference of the Parties (COP28) to the United Nations Framework Convention on Climate Change (UNFCCC) had several outcomes

- Transition away from fossil fuels: The conference agreed to transition away from fossil fuels, which was the first time a COP text mentioned a global shift away from fossil fuels.
- Accelerate action: The conference agreed to accelerate action across all areas by 2030.
- Triple renewable power: The conference agreed to triple renewable power by 2030.
- Double energy efficiency: The conference agreed to double energy efficiency by 2030.
- Operationalize the loss and damage fund: The conference operationalized the loss and damage fund on the first day of the conference.
- Recognize labor rights: The conference recognized labor rights in a climate decision for the first time.
- Highlight social protection: The conference highlighted social protection in regards to climate adaptation.
- End deforestation: The conference made international declarations on ending deforestation.
- Address health impacts: The conference made international declarations on the health impacts of climate change.

Introduction

- Air Composition
  - The air, generally unpolluted, contained 280 parts per million of carbon dioxide prior to the Industrial Revolution.
  - We currently have 380 parts per million of carbon dioxide in the air.

Gas in Atmosphere	Percentage
Nitrogen	78
Oxygen	21
Argon	1
Carbon Dioxide	0.038
Water Vapor	variable

Classification of pollutants

- Natural contaminants
  - Natural fog, pollen grains, bacteria and products of volcanic eruptions
- Aerosols
- Gases & vapors

No.	Group	Examples
1.	Sulphur compounds	SO_2, SO_3, H_2S, mercaptans
2.	Nitrogen compounds	NO, NO_2, NH_3
3.	Oxygen compounds	O_3, CO, CO_2
4.	Halogen compounds	HF, HCl
5.	Organic compounds	Aldehydes, hydrocarbons
6.	Radioactive compounds	Radioactive gases

Classification of pollutants

- Natural contaminants
  - Dust from places without vegetation
  - Methane gas emitted by the digestion of food by animals
  - Radon gas from radioactive decay
  - Wildfires, releasing smoke and carbon monoxide into the atmosphere
  - Volcanic activity, producing sulfur, chlorine, and ash particulates
  - Pollen grains
    - Pollen is technically a type of Particulate Matter but because most intact pollen grains are larger than 10 um, they don't usually fall into the PM10 category and are typically too large to enter our lungs.
      - *particulate matter is the only example that isn't a gas (PM 10, PM 2.5)*
    - Pollen grains can sometimes rupture or burst into smaller fragments. When that happens, the smaller fragments can become small enough to fall into the category of PM10 or PM2.5

Classification of pollutants

- Aerosol
  - Dispersion of solid or liquid particles of microscopic size in gaseous media, such as dust, smoke, or mist.
  - Differ widely in terms of particle size (0.01 μ -100 μ)
  - Dust
    - Produced by crushing, grinding, fly as from chimney, cement, foundry dust
    - Most of the dust particles settle to ground as dust fall, but particles 5 μ or small tend to form stable suspensions.
  - Smoke
    - Produced by incomplete combustion, size less than 1 μ
    - Size of coal particles : 0.2-0.01 μ and oil smoke particles 1-0.03 μ.
  - Mists
    - Low concentration of liquid particles in large size.
  - Fogs
    - Dispersion of water or ice in the atmosphere near the earth's surface reducing visibility to less than 0.5 km.
  - Fumes
    - Solid particles generated by condensation from the gaseous state, generally after Volatilization from melted substances.

Classification of pollutants

- Aerosol

Sources	Examples
1. Combustion	Fuel burning (coal, wood, fuel oil)
	Incineration (house and municipal garbage)
	Others (open fires, forest fires, tobacco smoking)
2. Materials handling and processing	Loading and unloading (sand, gravel, coal, ores, lime, cement)
	Crushing and grinding (ores, stone, cement, rocks, chemicals)
	Mixing and packaging (chemicals, fertilisers)
	Food processing (flour, corn starch, grains)
	Cutting and forming (saw mills, wall board, plastics)
	Metallurgical (foundries, smelters)
	Industrial (paper, textiles manufacture)
3. Earth-moving operations	Construction (road, buildings, dams, site clearance)
	Mining (blasting)
	Agriculture (soil filling, land preparation)
	Winds
4. Miscellaneous	House cleaning
	Mud road cleaning
	Crop spraying
	Poultry feeding
	Engine exhaust

Classification of pollutants

- Gases
  - Sulphur Dioxide
  - Hydrogen Sulphide
  - Hydrogen Fluoride
  - Chloride and Hydrogen Chloride
  - Oxides of Nitrogen
  - Carbon Mono oxide
  - Ozone
  - Aldehydes
  - Organic Vapor
  - Radioactive gases



Classification of pollutants

- Gases
  - Sulphur Dioxide
    - One of the principal constituents
    - Combustion of Coal
      - Anthracite less than 1 %
      - Bituminous around 4 %
    - Concentration in flue gases 0.05-0.25%.
    - Metallurgical operation: zinc, copper, and lead
      - During the smelting of theses ores sulphur dioxide is evolved in stack concentration of 5-10 %.

Classification of pollutants

- Gases
  - Hydrogen Sulphide
    - Foul smelling gas
    - Sources of its natural emission include anaerobic biological decay process on land, in marshes and in the oceans.
    - Volcanoes
    - Kraft pulp industry
    - Petroleum refineries
    - Coke-oven plant,
    - Chemical operations
  - Hydrogen Fluoride
    - Source: manufacture of phosphate fertilisers, aluminum industry, brick plants, ferro-enamels
    - Concentration: 0.001-0.1 ppm



Classification of pollutants

- Gases
  - Chlorine and hydrogen chloride
    - Manufacturing of other chemicals
    - Water purification
    - Equipment failures
    - Main effects: respiratory irritation from chlorine, corrosion by hydrogen chloride, and damage to vegetation.
  - Oxides of Nitrogen
    - Source: gaseous emissions from where nitric acid is produced.
    - Automobile exhaust
    - Nitric oxide and nitrogen dioxide arises from human activity
    - NOx
    - Ammonia and ammonium salts are not important air contaminants

Classification of pollutants

- Gases
  - Carbon monoxide
    - Odourless and colourless
    - Incomplete combustion of carbonaceous materials
    - Highly poisonous gas
    - Automobile exhaust is the main source.
  - Ozone
    - Combustion and sunlight
    - Highly poisonous and smelly

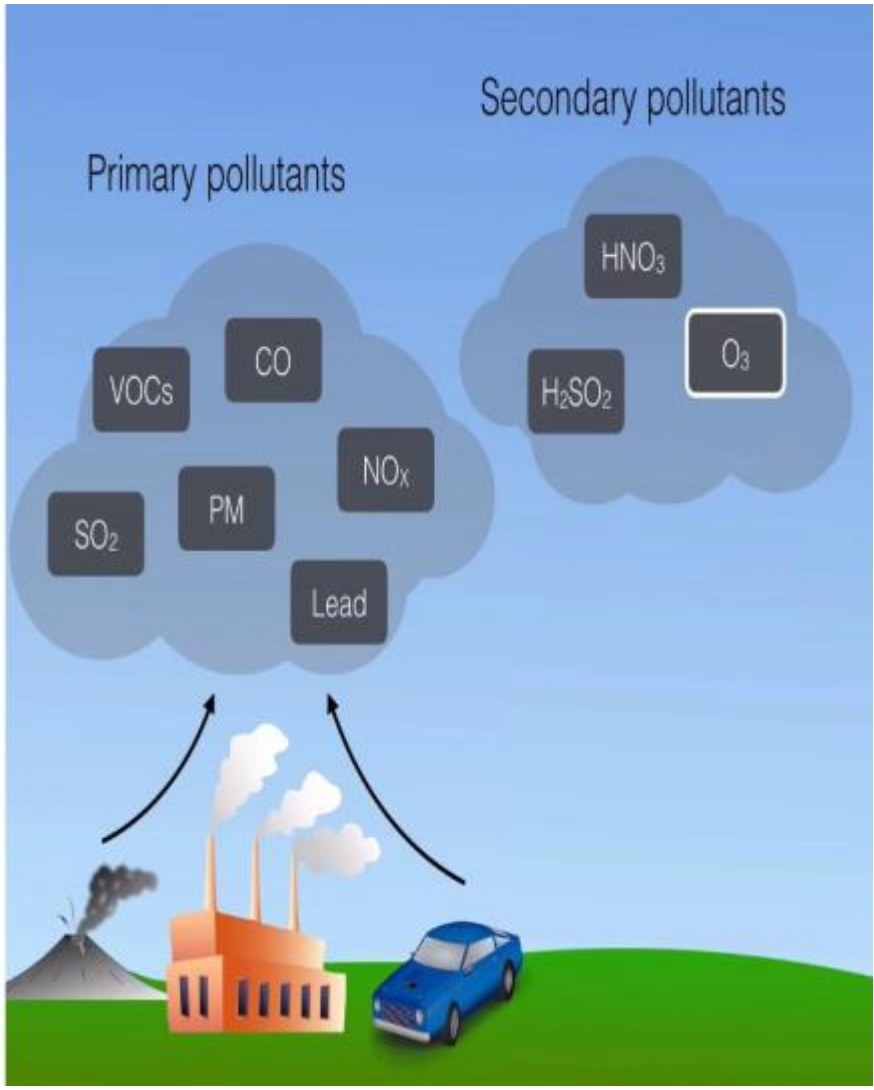
Classification of pollutants

- Gases

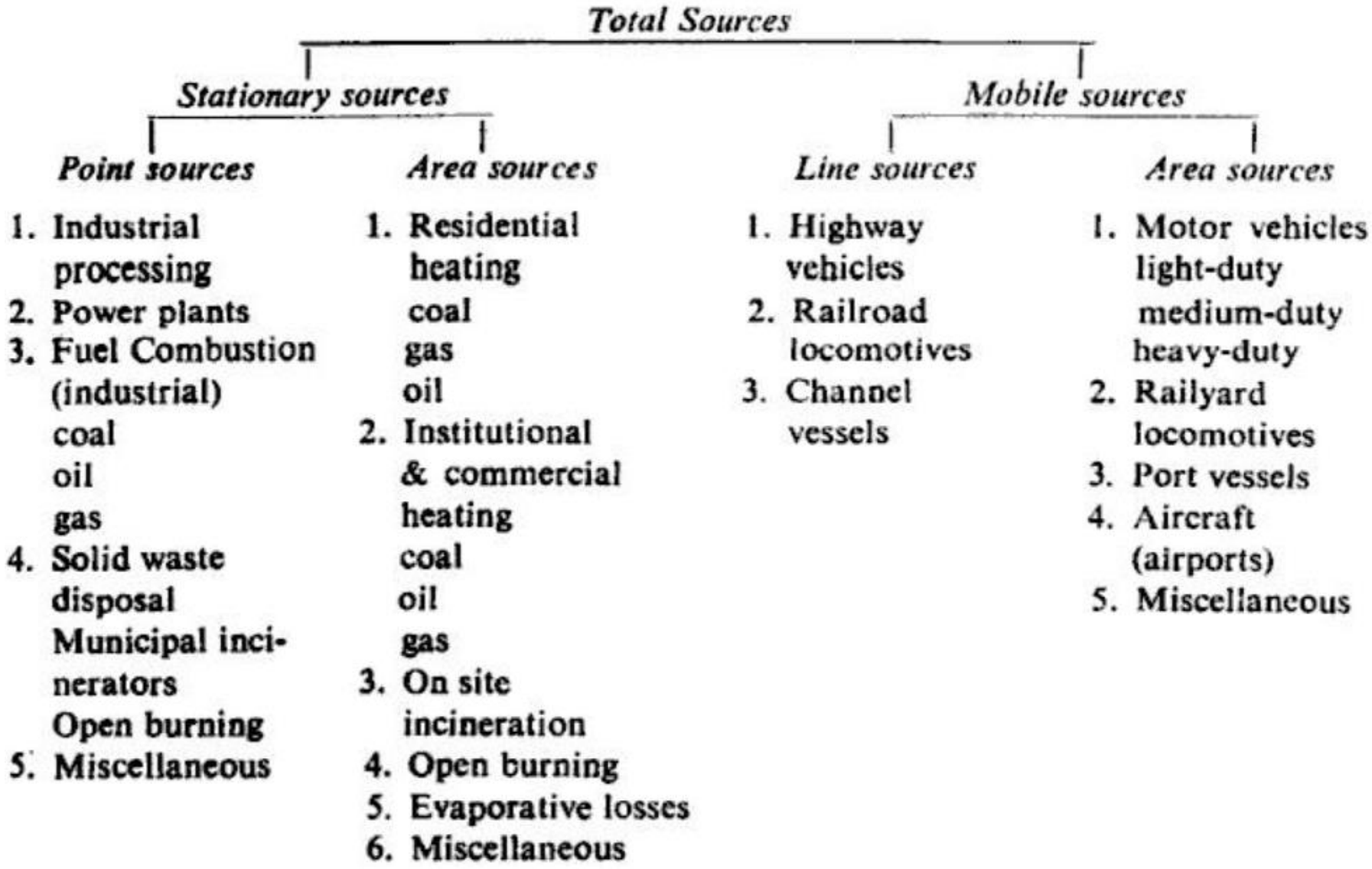
No.	Class	Aerosols	Gases and vapours
1.	Combustion processes (domestic burning, thermal power plants, cars, trucks, aeroplanes, and railways. Also refuse burning)	Dust, fume, smoke	SO_2, NO_2, CO, organic vapours, odours
2.	Chemical processes (paper mills, cement, fertilisers, etc.)	Dust, fume, mist	Process-dependent (SO_2, CO, NH_3, NO_2, organic vapours, odours)
3.	Petroleum operations	Dust, mist	SO_2, H_2S, NH_3, CO, hydro-carbons, mercaptans
4.	Metallurgical processes (aluminium refineries, steel plants)	Dust, fume	SO_2, CO, fluorides, organic vapours
5.	Mineral processing	Dust, fume	Process-dependent (SO_2, CO, fluorides, organic vapours)
6.	Food and feed operations	Dust, mist	Odourous materials
7.	Agricultural activities		Organic phosphates, chlorinated hydro-carbons sulphur oxides, organic vapours
	(a) Crop spraying	Dust, mist	
	(b) Field burning	Smoke, fly ash	
8.	Nuclear energy programmes		
	(a) Fuel fabrication		Fluorides
	(b) Ore preparation	Dust	Iodine-131 and Argon-41
	(c) Bomb explosion		Radioactive gases (Sr-90, Cs-137, C-14, etc.)

Classification of pollutants

- Primary Pollutants
  - Volatile organic compounds (VOCs)
  - Carbon monoxide (CO)
  - NOx
  - Sulfur dioxide (SO2)
  - Particulate matter (PM)
  - Lead
- Secondary Pollutants
  - Ozone: Chemical reaction between nitrogen dioxide, VOCs and sunlight.
  - Acid Mists (H2SO4): Reaction of sulphur dioxide and dissolved oxygen
  - Nitric Acid: Due to NOx



Sources of Air Pollutions



Effects of Air Pollution

- Effects on Human Health
- Effects on Animals
- Economic Effects

Effects of Air Pollution on Human Health

- The prime factors affecting human health
  - Nature of the pollutants
  - Concentration of the pollutants
  - Duration of exposure
  - State of health of the receptor
  - Age group of receptor

Air Pollution episodes in the world

S. No.	Month and year	Place	Mortality
1.	December, 1930	Meuse Valley (Belgium)	63
2.	October, 1948	Donora (Pennsylvania)	20
3.	November, 1950	Poza Rica (Mexico)	22
4.	December 1952	London	4000
5.	November, 1953	New York	220
6.	January, 1956	London	1000
7.	December, 1957	London	750
8.	December, 1962	London	700
9.	January, 1963	New York	300
10.	November, 1966	New York	168
11.	December, 1984	Bhopal (India)	2000

Effects of Air Pollution on Human Health

High air pollution levels can cause immediate health problems including:	Aggravated cardiovascular and respiratory illness
	Added stress to heart and lungs, which must work harder to supply the body with oxygen
	Damaged cells in the respiratory system

Effects of Air Pollution on Human Health

Long-term exposure to polluted air can have permanent health effects such as:

- Accelerated aging of the lungs
- Loss of lung capacity and decreased lung function
- Development of diseases such as asthma, bronchitis, emphysema, and possibly cancer
- Shortened life span

Effects of Air Pollution on Human Health

Short-term exposure to particulate pollution can:

- Aggravate lung disease causing asthma attacks and acute bronchitis
- Increase susceptibility to respiratory infections
- Cause heart attacks and arrhythmias in people with heart disease

Even if you are healthy, you may experience temporary symptoms, such as:

- Irritation of the eyes, nose and throat
- Coughing
- Chest tightness
- Shortness of breath

Effects of Air Pollution on Human Health

Those most susceptible to severe health problems from air pollution are:

- Individuals with heart disease, coronary artery disease or congestive heart failure
- Individuals with lung diseases such as asthma, emphysema or chronic obstructive pulmonary disease (COPD)
- Pregnant women
- Outdoor workers
- Older adults and the elderly
- Children under age 14
- Athletes who exercise vigorously outdoors

Effects of Air Pollution on Human Health

Health Effects from Specific Pollutants

Ground-level Ozone

It can also cause other health problems including:

- Aggravated respiratory disease such as emphysema, bronchitis and asthma
- Lung damage, even after symptoms such as coughing, or a sore throat disappear
- Wheezing, chest pain, dry throat, headache or nausea
- Reduced resistance to infections
- Increased fatigue
- Weakened athletic performance

Effects of Air Pollution on Human Health

Long-term exposure to particulate pollution can result in significant health problems including:

- Increased respiratory symptoms, such as irritation of the airways, coughing or difficulty breathing
- Decreased lung function
- Aggravated asthma
- Development of chronic respiratory disease in children
- Development of chronic bronchitis or chronic obstructive lung disease
- Irregular heartbeat
- Nonfatal heart attacks
- Premature death in people with heart or lung disease, including death from lung cancer

Effects of Air Pollution on Human Health

- Health Effects
  - Eye Irritation
  - Nose and throat irritation
  - Irritation of the respiratory tract
  - Morbidity and mortality rate
  - Asthmatic attacks
  - Chronic pulmonary disease
  - Carbon monoxide combines with haemoglobin in blood and consequently increases stress
  - Hydrogen fluoride cause disease of the bone
  - Carcinogenic agents cause cancer
  - Dust Particles cause respiratory disease
  - Certain heavy metals like lead may cause poisoning

Effects of Air Pollution on Animal

- Farm Animal (fluorine, arsenic, lead)
  - Accumulation of air-borne contamination in the vegetation.
  - Subsequent poisoning of the animal when they eat the contaminated vegetation.

- Pets (Asthma, suffering from bronchitis, lack of appetite)
  - Reduction in life span
  - Cancer
  - Genetic effects

Economics Effects

- Air pollution is detrimental not only for the environment and its life forms but also to economic growth.
- The total market costs due to air pollution include both direct and indirect costs.
  - Direct cost: changes in value generated in different sectors from changes in labour productivity, increased health expenditures and changes in value generated in agriculture from changes in crop yields.
  - Indirect cost: relocation of factors of production, and changes in international trade and savings.

❑ The effect of air pollution on the economy may be a derived one. In simple language, the economy thrives when people are healthy, and business that depends on cultivated raw materials and natural resources are running at full efficiency.



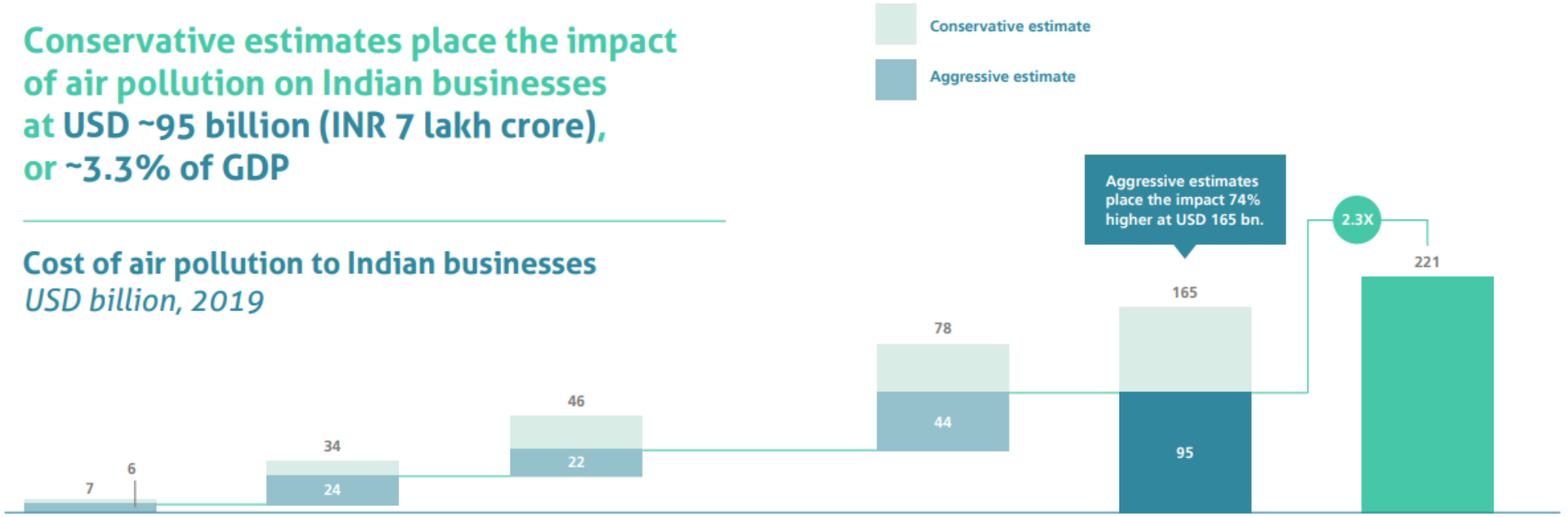
Economics Effects

- Air pollution costs Indian businesses USD 95 billion or 3% of India's GDP every year.
- Air pollution impacts businesses along three direct impact pathways. These are (a) the opportunity cost of lost worker productivity (e.g., due to higher absenteeism), (b) revenue lost due to reduced consumer footfall, and (c) the opportunity cost of premature mortality attributed to air pollution.
- Due to air pollution caused early deaths estimated that USD225 billion in global income was lost annually; these losses were highest in South Asia, where they amounted to 1% of regional GDP.
- India's employers lose ~1.3 billion working days per year to air pollution, amounting to an annual loss of USD ~6 bn

Economics Effects

Conservative estimates place the impact of air pollution on Indian businesses at USD ~95 billion (INR 7 lakh crore), or ~3.3% of GDP

Cost of air pollution to Indian businesses
USD billion, 2019



Absenteeism

~1.3 billion working days are lost p.a. due to absenteeism

Absenteeism impact is higher with working age population with dependents, such as children

Presenteeism

Presenteeism is a hidden cost which may directly impact corporate profits

Skilled services and manufacturing sector are disproportionately affected

Consumer footfall

Air pollution drives a 1.3% decrease in India's consumer spending

The impact in segments such as international tourism is as high as 7.41%

Premature Mortality[1]

India had 1.7 million premature deaths from air pollution in 2019,1~18% of overall mortality in the same year

India contributes ~25% to the global cost of premature mortality

Total impact

The total economic impact of air pollution is equal to:

150% India's healthcare budget

150% India's defense spending

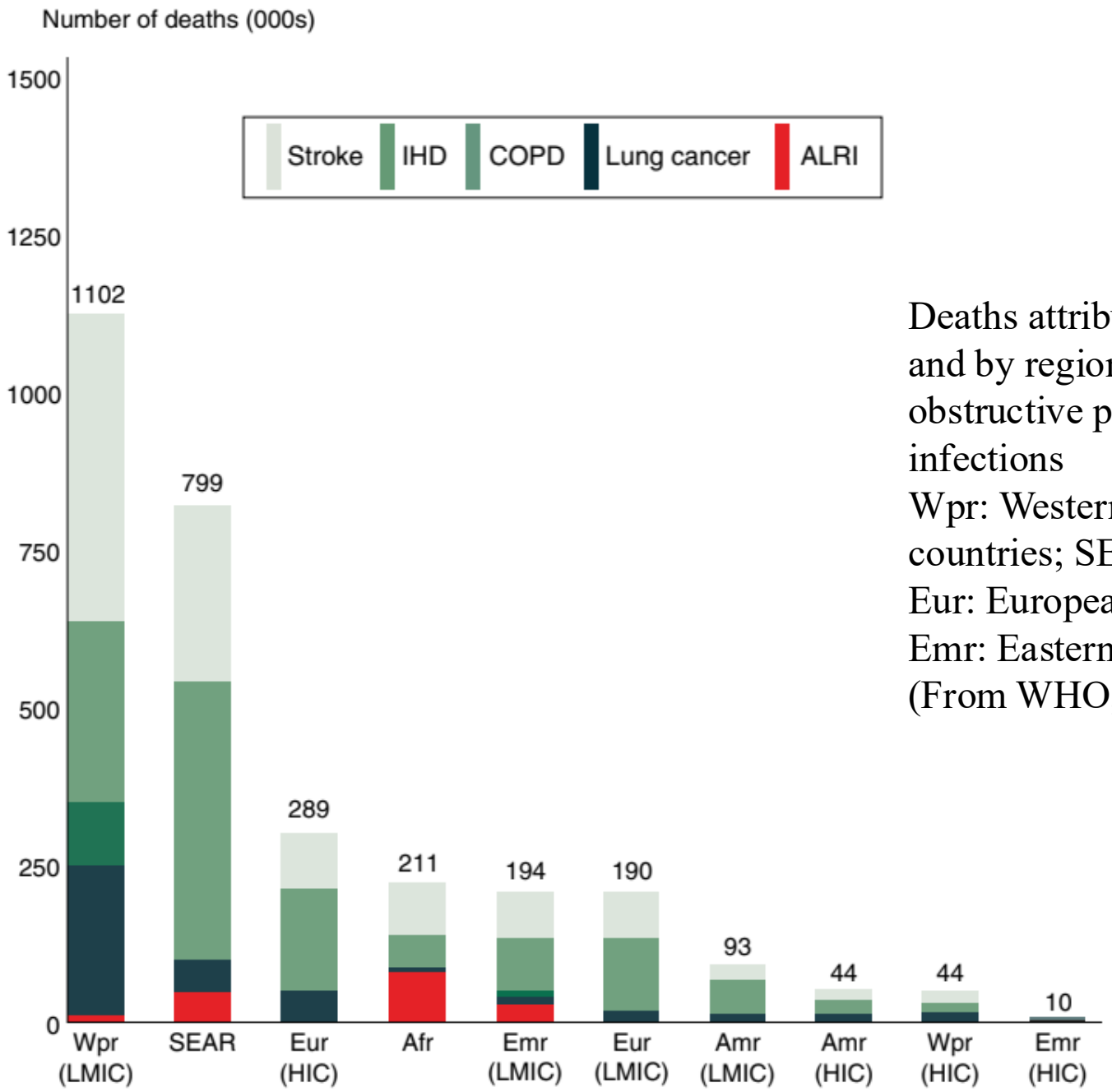
50% of tax collected annually

Loss due to COVID[2]

Cost of air pollution to businesses is equal to 43% of the pandemic's impact

Unlike the pandemic, this is an annually recurring cost

Despite this, only INR 4,400 cr. has been invested in air pollution, <10% of total outlay on COVID response



Deaths attributable to ambient air pollution in 2012, by disease and by region (IHD: ischaemic heart disease; COPD: chronic obstructive pulmonary disease; ALRI: acute lower respiratory infections
Wpr: Western Pacific region; LMIC: low- and middle-income countries; SEAR: South East Asian Region;
Eur: European Union; HIC: high-income countries; Afr: Africa; Emr: Eastern Mediterranean; Amr:America).
(From WHO, 2016a.)

Air quality index (AQI)

- National Air Quality Index (AQI) transforms complex air quality data of eight pollutants into a single number (index value), nomenclature and colour.

AQI Category, Pollutants and Health Breakpoints								
AQI Category (Range)↓	→ Categories for the various readings of the pollutant based on the health breakpoints or health impacts							
	PM_{10} 24-hr	$PM_{2.5}$ 24-hr	NO_2 24-hr	O_3 8-hr	CO 8-hr (mg/m^3)	SO_2 24-hr	NH_3 24-hr	Pb 24-hr
Good (0-50)	0-50	0-30	0-40	0-50	0-1.0	0-40	0-200	0-0.5
Satisfactory (51-100)	51-100	31-60	41-80	51-100	1.1-2.0	41-80	201-400	0.5 –1.0
Moderately polluted (101-200)	101-250	61-90	81-180	101-168	2.1- 10	81-380	401-800	1.1-2.0
Poor (201-300)	251-350	91-120	181-280	169-208	10-17	381-800	801-1200	2.1-3.0
Very poor (301-400)	351-430	121-250	281-400	209-748*	17-34	801-1600	1200-1800	3.1-3.5
Severe (401-500)	430 +	250+	400+	748+*	34+	1600+	1800+	3.5+

- *AQI is considered as 'One Number- One Colour-One Description' for the common man to judge the air quality within his vicinity. The formulation of the index was an initiative under Swachh Bharat Mission (Cleanliness Mission), based on the recommendations of IIT Kanpur and the Expert Group formed in this regard.*

Air quality index (AQI)

- The AQI Index values and their associated health impacts are as follows:

AQI	Associated health impacts	Colour code
Good (0–50)	Minimal impact	
Satisfactory (51–100)	May cause minor breathing discomfort to sensitive people	
Moderately polluted (101–200)	May cause breathing discomfort to people with lung disease such as asthma, and discomfort to people with heart disease, children and older adults	
Poor (201–300)	May cause breathing discomfort to people with prolonged exposure, and discomfort to people with heart disease	
Very poor (301–400)	May cause respiratory illness to people with prolonged exposure. Effect may be more pronounced in people with lung and heart diseases	
Severe (401–500)	May cause respiratory impact even on healthy people, and serious health impacts on people with lung/heart disease. The health impacts may be experienced even during light physical activity	

Cost-effective technologies to curb air pollution

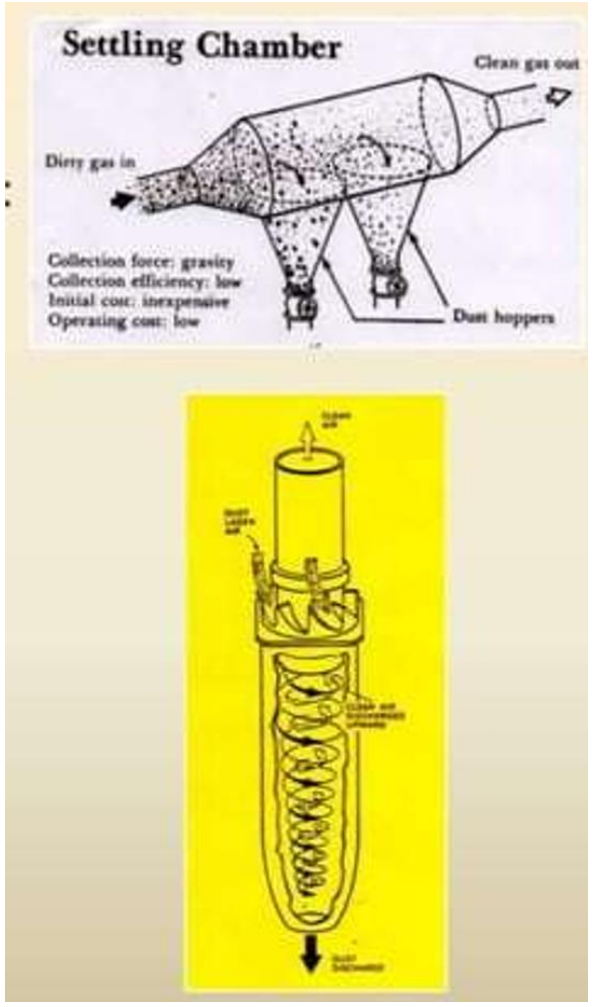
Pollutant	Main source
Suspended particulate matter (SPM)	Power plants, automobiles, boilers, cement industries
Sulfur oxides (SO_2)	Power plants, boilers, sulfuric acid manufacture, petroleum refining
Nitrogen oxides (NO_2)	Automobiles, power plants, nitric acid manufacturing
Carbon monoxide (CO)	Automobiles
Hydrogen sulfide (H_2S)	Pulp and paper, petroleum refining
Hydrocarbon (HC)	Automobile, petroleum refining
Ammonia (NH_3)	Fertilizer plant
Lead	Automobile, battery manufacturing

Elimination approach

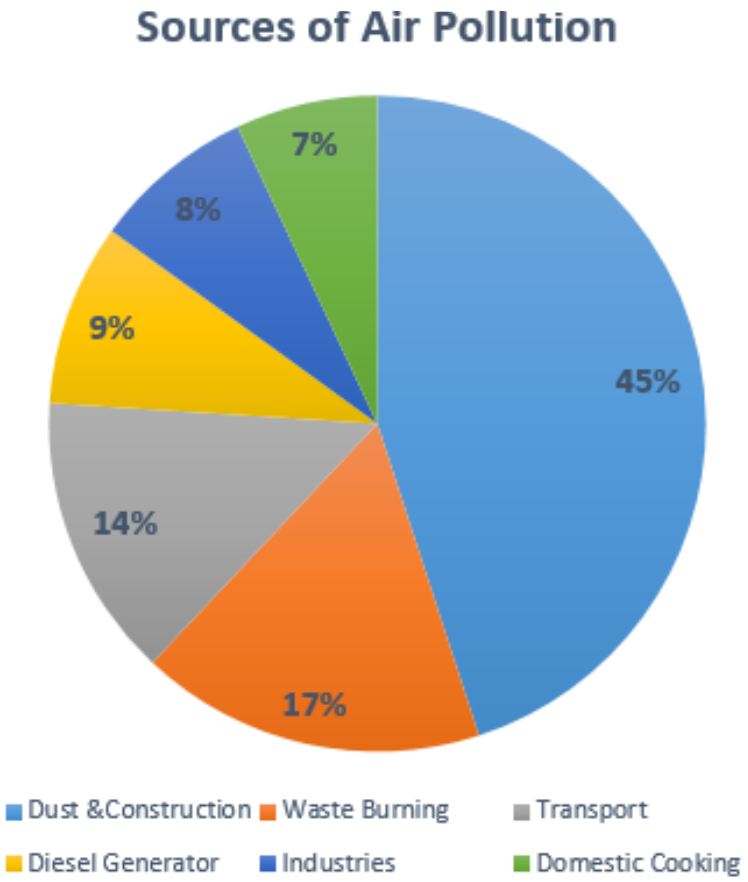
- Proper fuel-firing practices allowing combustion improvement, along with adequate air supply, can help in lowering incomplete combustion products.
- Choice of fuels, Fuel cleaning, Advance technology for combustion

Emission-control approaches

- Strategies towards controlling emissions of air pollutants are important in protecting public health and welfare.
- Particle-control system
    - Gravity settling chambers
    - Cyclones
    - Electrostatic precipitators
- Gas and vapour phase control systems
    - Wet scrubbers
    - Activated charcoal



Air Pollution Control: Policies & Legislations



Industrial activities and ever-increasing number of vehicles and population has made India the biggest contributor of air pollution among South Asian countries.

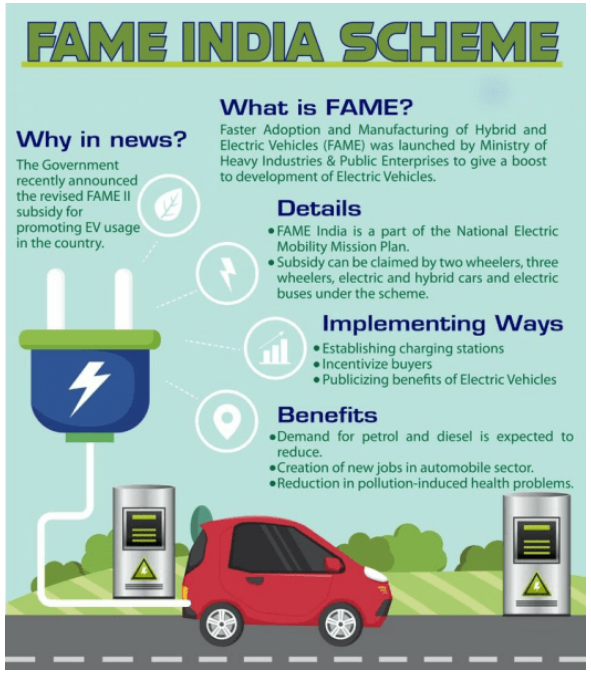
According to the WHO, Of the 30 most polluted cities in the world, 21 were in India in 2019.

A remarkable effort made by India in 1976 was in amending its constitution (the 42nd amendment) to provide constitutional provisions to protect the environment under section 48A and 51A (g).

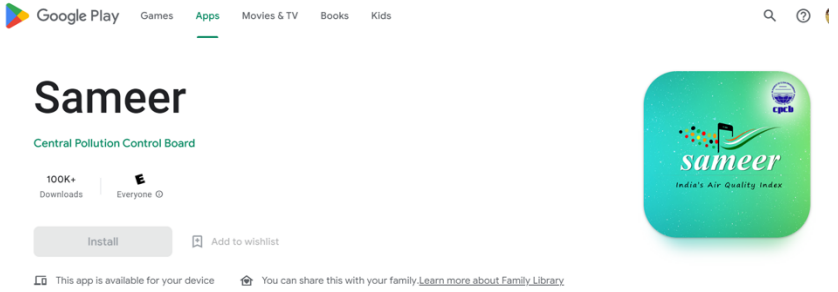
Article 48A, the directive principle of state policy, mandates the state ‘to protect and improve the environment and safeguard the forests and wildlife of the country’.

Article 51A (g) imposes a fundamental duty and a social obligation on every citizen ‘to protect and improve the natural environment including forests, lakes, rivers and wildlife and to have a compassion for living creatures

Air Pollution Control: Policies & Legislations



- Air (Prevention and Control of Pollution) Act 1981, also known as the Air Act.
- At present, each sector has regulatory measures set by the policy makers of India. (Ex: **National Clean Air Programme, Centre-run System of Air Quality and Weather Forecasting and Research (SAFAR))**
- Ministry of Road Transport and Highways (MoRTH) jumped from BS-IV to BS-VI emission standards.
- Government of India is also taking incentives for the manufacturing and promotion of electric vehicles.
- Policy makers are promoting investment in renewable energy, for example, the National Solar Mission.
- Encouragement for purchasing energy efficient and pollution-free technologies
- The National Green tribunal has announced a ban on open burning of rubbish and directed all the concerned authorities for the regulation of the same .



Montreal Protocol

In 1985 and 1987, new research showed that 40% of the ozone layer had been depleted over Antarctica between 1957 and 1984, and a hole had grown to the size of the United States.

This report led to the formation of international negotiations between the political leaders of the world, known as the Montreal Protocol, signed on 8 September 1987 at the Vienna convention.

50% reduction in chlorofluorocarbons (CFCs) by1998.

Developing nations to increase emissions up to specified limit to meet ‘basic domestic needs’ but for a limited period of 10 years, subject to a reduction of 50% for the next 10 years.

London agreed to phase out 15 different CFCs worldwide, the European Community announced the phasing out of CFCs by 1997 and a total elimination of ‘use and production’ of CFCs by 2000.

Montreal Protocol

What does the Montreal Protocol do?

- Bans the production and import of ODS
- Reduces the concentration of ODS in the atmosphere
- Protects the ozone layer
- Helps protect the Earth's climate system
- Saves millions of people from cancer

Kyoto Protocol

- Adopted in 1997, in Japan as a result of the United Nations Framework Convention on Climate Change UNFCCC of 1992 held in Rio de Janeiro; the famous Earth Summit.
- Annex I nations would reduce emissions by 5% and these targets were to be achieved between 2008 and 2012.
- USA at 7% reduction emissions, EU at 8 % and Japan at 6% .
- Countries such as China and India were among the largest emitters of GHGs, but they were not controlled by the Kyoto Protocol.

What greenhouse gases were targeted?

Carbon dioxide, Methane, Nitrous oxide, Hydrofluorocarbons, Perfluorocarbons, and Sulphur hexafluoride

?